

Mariana Pachón Puentes

Creative Technologist · Designer Engineer

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INSTALLATIONS

Flora, Electronic Garden

Interactive Installation · 2025

- Designed and built an interactive ecosystem simulating plant and animal life in crisis through 3D physical objects, sensors, and actuators.
- Engineered the hardware system: 4 Arduinos coordinating 6 sensors (moisture, RFID) and 9 actuators (water pumps, lighting, motion) to simulate organic behaviors like blooming and growth.
- Led hardware and software design in a 4-person team, from concept through fabrication and exhibition.

Explorer Award, A MAZE Festival · Featured on the Arduino Blog · BCN Games Fest · Berlín & Barcelona

Wear the Sound

Interactive Installation · 2025

- Designed and built haptic wearables and reclining supports letting visitors feel live music through vibration, expanding festival accessibility.
- Engineered wearable hardware to receive real-time OSC signals from a live music console, syncing vibration to a collaborating artist's mix.
- Adapted the physical design to the architecture of the Mies van der Rohe Pavilion exhibition space.

Sónar +D Festival, Mies van der Rohe Pavilion · BCN Games Fest · Barcelona

Memorias desbloqueadas en el trópico

Ephemeral Installation · 2025

- Created an ephemeral installation exploring shared bodily memory between Colombia and Indonesia through a map printed on banana leaf and tropical fruit.
- Built the electronic component: a fluorescent light triggered by a proximity sensor when visitors reach for fruit, activating the connection between the two countries on the map.
- Printed the map on banana leaf using a BrailleRAP machine, connecting the materiality of the tropics to the concept.

DINACON Residency · Les, Bali, Indonesia

EXPERIMENTS & PROTOTYPES

Burbúfonos

ESP32-based phone system that converts voice into bubble signals via two hacked bubble machines and two microphones.

Noise boxes

Sound experiments combining metal, motors, and piezoelectric sensors to generate physical noise.

Shrinking theatre

Mechanical stage prototype using dual rack-and-pinion mechanisms driven by continuous rotation servos.

EDUCATION

M.A. in Audiovisual Innovation & Interactive Environments

BAU, College of Arts & Design — Barcelona, Spain (2023–2024)

B.S. in Computer Science — High Distinction, GPA 3.75/4.0

Worcester Polytechnic Institute — USA (2016–2019)

SOFTWARE & PRODUCT EXPERIENCE

Frontend Engineer — TaxGuardian (Freelance)

Nov 2025 – Mar 2026

- Built customer and admin interfaces for a tax services product (React, TypeScript, Vite). Established CI/CD pipelines on Azure and component library with Storybook.

Fullstack Engineer — Healthcasts AI / Travoya.ai (Freelance)

2024–2025

- Shipped end-to-end features across two early-stage AI products: medical case summaries (Python, Svelte) and a video processing pipeline to extract travel recommendations from social media.

Fullstack Engineer — MojoTech (Agency)

2020–2023 · Boulder, CO

- Built and shipped product features (React, Rails) for client teams; optimized a provider onboarding flow cutting costs by \$4.8M and improving retention 7%.

JavaScript Instructor — Codeworks

Nov 2023 – Jan 2024 · Barcelona

- Mentored bootcamp students through project-based learning in React, Next.js, and Express.

SKILLS

Fabrication & Hardware

Arduino, ESP32, sensors & actuators, electronics, 3D printing, laser cutting, CNC, servo/motor mechanisms

Programming

React, TypeScript, Python, Ruby on Rails, Svelte, SQL, Git, Processing

Creative

Ableton Live, OSC, sketching & prototyping, Storybook

Languages

Spanish (Native) · English (Fluent) · French (Fluent)